

Exploring the Capabilities of Multimodal Large Language Model Through Multimodal Misinformation Detection

Abstract

Multimodal large language Model (MLLMs) have emerged as transformative tools in addressing diverse tasks by integrating and processing information from multiple modalities. Their ability to understand and correlate cross-modal information enables more nuanced and context-aware decision-making, which is crucial for solving complex real-world problems. From enabling multimodal sentiment analysis and misinformation detection to advancing applications in healthcare, education, and human-computer interaction, MLLMs are redefining the boundaries of AI capabilities. By leveraging their pre-trained knowledge across modalities, MLLMs demonstrate superior performance and adaptability, making them indispensable for addressing the challenges posed by diverse and multimodal data environments. This paper explores the principles and current advancements of Multimodal Large Language Model, emphasizing their transformative potential in understanding and integrating diverse modalities. We provide an overview of existing models, highlighting their architectures, training strategies, and applications across various domains. To demonstrate their capabilities, we delve into a specific task—multimodal misinformation detection. This task requires models to analyze and correlate information from multiple modalities, such as text and images, to identify and flag misleading or false content. By focusing on this application, we aim to showcase the strengths and limitations of Multimodal large language Model.

Keywords

Multimodal large language Model, Fundamentals, Capabilities, Multimodal misinformation detection

1 Introduction

Multimodal Large Language Models (MLLMs) represent a significant leap in the field of artificial intelligence, offering a sophisticated approach to processing and understanding data from multiple modalities, including text, images, audio, and video. These models possess the unique capability of integrating and correlating information from diverse sources, enabling them to perform tasks that were once considered highly complex or even unattainable for traditional single-modality models. By combining the strengths of different data types, MLLMs facilitate more accurate and context-aware decision making, making them invaluable for addressing a wide range of real-world challenges.

The application of MLLMs spans various domains, from sentiment analysis and misinformation detection to healthcare, education, and human-computer interaction. In particular, MLLMs are proving to be game changers in the fight against misinformation, where the ability to analyze and interpret both textual and visual content is essential for identifying misleading or false narratives. By leveraging their pre-trained knowledge across modalities, MLLMs offer superior performance and adaptability, making them indispensable in today's multimodal data environments.

This paper provides a detailed exploration of the principles, architectures, and training strategies behind MLLMs, with a particular focus on their transformative potential to understand and integrate various types of information. We highlight current advancements in this area, examining existing models and their applications across various domains. To illustrate the capabilities of MLLMs in practice, we focus on the specific challenge of multimodal misinformation detection, an area that requires the fusion of textual and visual data to effectively identify and mitigate false or misleading content. Through this case study, our aim is to demonstrate both the strengths and limitations of MLLMs, providing insights into their real-world applicability and future prospects.

2 Fundamentals of Multimodal Large Language Models

Multimodal Large Language Models represent a significant leap in artificial intelligence by integrating and processing information across diverse modalities. These models leverage the power of transformer architectures, large-scale pre-training, and fine-tuning to enable cross-modal understanding and reasoning. By aligning features from different modalities, MLLMs create a unified representation that enhances their ability to perform a variety of tasks requiring nuanced and context-aware decision-making. This section provides an in-depth exploration of the foundational concepts underpinning MLLMs, including their reliance on transformer architectures, the role of pre-training and fine-tuning, strategies for cross-modal alignment, and the specialized encoders that enable efficient and accurate multimodal processing.

2.1 Transformer architecture

The transformer architecture has become the backbone of Multimodal Large Language Models (MLLMs), enabling efficient and scalable processing of multimodal data. Transformers are designed to handle sequential data with unparalleled parallelism and flexibility. The architecture's modularity and ability to model long-range dependencies make it ideally suited for integrating and reasoning across diverse modalities such as text, images, audio, and video. Fig 1 shows the components of transformer. Key Components of the Transformer Architecture are:

Positional Encoding: Positional encoding is a technique used to inject information about the position of tokens within the input sequence. Since the self-attention mechanism in Transformers operates in parallel across the entire sequence and does not inherently capture token order, positional encoding is essential for maintaining the sequential nature of the data. Positional encodings are added to the input embeddings at the bottom of the model, allowing the network to distinguish between tokens based on their positions in the sequence. The original Transformer architecture, as proposed by Vaswani et al., employs sinusoidal positional encoding. In contrast to sinusoidal positional encodings, learned positional embeddings are another approach, where the positional encodings

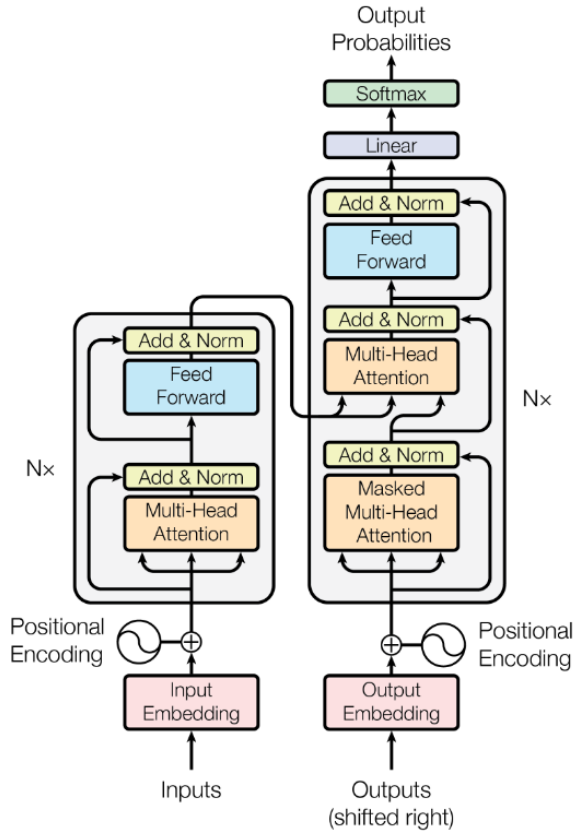


Figure 1: Transformer Architecture (Adapted from Vaswani et al., 2017)

are treated as parameters to be learned during training. This method allows the model to potentially discover more optimal positional representations based on the task at hand.

Multi-Head Self-Attention: The self-attention mechanism, which is the foundation of transformers, involves computing attention scores between all pairs of tokens in the input sequence. These scores are used to create weighted combinations of the input tokens, allowing the model to capture contextual dependencies at varying distances. The multi-head self-attention mechanism allows the model to compute multiple attention scores simultaneously, capturing diverse aspects of relationships within the data.

Feedforward Layers: The feedforward layer is a component that follows the self-attention mechanism. Typically, the feedforward layer consists of two fully connected layers: the first expands the input dimension, while the second projects it back to the original dimension. This structure enables the model to learn more complex nonlinear transformations, thereby enhancing its expressive power. The workflow of the feedforward layer can be described as follows: the input is first mapped to a higher-dimensional space via a linear transformation, followed by a nonlinear activation function (usually ReLU or GELU); then, a second linear transformation maps the high-dimensional representation back to the original dimension.

Unlike the self-attention mechanism, the feedforward layer operates independently at each position, meaning it does not consider the dependencies between different positions in the input sequence. The feedforward layer also includes residual connections and layer normalization, which help mitigate the vanishing gradient problem and improve both training efficiency and stability.

Layer Normalization and Residual Connections: Layer normalization and residual connections contribute to stable training, effective gradient flow, and enhanced model performance. These techniques are particularly important in deep neural networks, where issues such as vanishing gradients and overfitting can hinder optimization. Layer normalization works at the level of individual samples. This makes it particularly suitable for sequence modeling tasks, as it avoids dependencies on batch statistics and maintains consistency across varying sequence lengths. Residual connections address the problem of vanishing gradients in deep networks by providing a direct pathway for the gradient to flow during back-propagation. This improves gradient stability and allows for the training of much deeper networks. Additionally, residual connections make it easier for the network to learn identity mappings, ensuring that uninformative layers do not degrade performance.

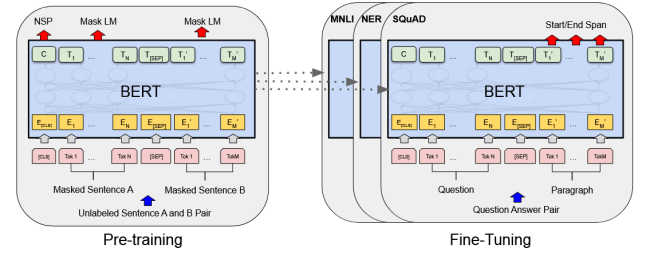


Figure 2: Pre-training and Fine-tuning for BERT, a language model in 2019, which indicates the start of pre-training and fine-tuning for the language model.(Adapted from Devlin et al., 2019)

2.2 Pre-training

Pre-training serves as the foundation of Multimodal Large Language Models (MLLMs), enabling them to acquire generalizable knowledge from large-scale multimodal datasets. This phase involves training the model on diverse tasks and objectives to learn representations that capture the relationships and alignments between different modalities, such as text and images, audio and video, or combinations thereof. By leveraging extensive pre-training, MLLMs gain the ability to understand and reason across modalities, which can then be fine-tuned for specific downstream applications.

During pre-training, MLLMs finish Cross-Modal Alignment, Representation Learning and Task Generalization. The models align features from different modalities into a shared representation space, learn rich representations of each modality independently and in combination and develop general-purpose capabilities that allow them to adapt to new tasks with minimal fine-tuning.

2.3 Fine-tuning for specific tasks

Pre-training provides the model with generalized knowledge from large-scale, multimodal data, fine-tuning enables the model to specialize in particular tasks or domains by adjusting its parameters based on task-specific labeled data. Fine-tuning ensures that the knowledge learned during pre-training is effectively transferred to real-world applications, enhancing the model's performance on tasks. While pre-training establishes a general understanding of the relationships between different modalities, fine-tuning improves the model's ability to reason across modalities for particular tasks.

In End-to-End Fine-Tuning, all the layers of the pre-trained model are updated during the fine-tuning process. The model adapts to the task by adjusting the weights of the entire architecture, allowing it to learn task-specific features and patterns. In Layer-wise Fine-Tuning, fine-tuning is applied selectively to certain layers of the model, rather than the entire architecture. For example, earlier layers may focus on general representations (e.g., low-level visual features or syntactic structure in text), while later layers are fine-tuned for task-specific knowledge (e.g., sentiment classification or misinformation detection). This approach can be more computationally efficient, especially for large models.

2.4 Efficiency optimizations

As Multimodal Large Language Models grow in size and complexity, training and deploying these models come with significant computational challenges. The high resource demands, including memory, processing power, and energy consumption, make it difficult to scale these models for real-world applications. Efficiency optimizations are essential for overcoming these challenges while maintaining model performance. These optimizations aim to reduce the computational cost, memory footprint, and inference latency of MLLMs without sacrificing their accuracy.

Pruning: Pruning is a technique where certain weights or connections in the model are removed (set to zero), reducing the overall size of the model without significantly affecting its performance. Pruning can be done either by removing entire neurons or by zeroing out weights based on their importance. Techniques like structured pruning (removing entire filters or layers) and unstructured pruning (removing individual weights) are commonly used in MLLMs to reduce both memory and computational requirements.

Quantization: Quantization reduces the precision of the weights, typically from 32-bit floating point to lower bit-width formats like 16-bit or even 8-bit integers. This process significantly reduces the model's memory usage and speeds up inference by allowing operations to be performed using more efficient arithmetic. Quantization has become especially important for deploying large models on resource-constrained hardware, such as mobile devices or edge computing platforms.

LoRA: LoRA (Low-Rank Adaptation) is a technique that allows large pre-trained models to be fine-tuned efficiently by introducing a low-rank decomposition into the weight matrices of the model, without needing to update the entire set of parameters. Rather than directly fine-tuning the model's original weight matrices, LoRA modifies only a small subset of parameters that are represented in low-rank form. This adaptation reduces the number of trainable parameters, thus lowering memory and computational costs while

maintaining high performance. LoRA is particularly effective in large transformer-based models, where the number of parameters is vast, and full fine-tuning is often computationally prohibitive. By focusing on adapting a low-rank subspace of the model's weight matrices, LoRA enables efficient task-specific fine-tuning without the need to retrain or store the entire set of original parameters.

Knowledge Distillation: Knowledge distillation involves transferring knowledge from a large, complex model (the teacher) to a smaller, more efficient model (the student). The student model is trained to mimic the behavior of the teacher by minimizing the difference in predictions, allowing it to achieve similar performance with fewer parameters.

Mixed Precision Training: Mixed precision training leverages both 16-bit and 32-bit floating-point arithmetic to accelerate model training while maintaining computational efficiency and numerical stability. The main idea behind it is to use lower precision (16-bit) for the majority of operations, such as matrix multiplications and gradient calculations, and reserve higher precision (32-bit) for critical operations, like the accumulation of gradients and weight updates. This approach significantly reduces memory usage and improves the throughput of the training process without sacrificing model accuracy. Modern GPUs, such as NVIDIA's Volta and Ampere architectures, are specifically designed to exploit the computational advantages of mixed precision, delivering higher performance compared to traditional single-precision training. However, to maintain model accuracy with reduced precision, various strategies such as loss scaling are employed. Loss scaling involves multiplying the loss by a constant factor before performing the backpropagation step, which helps to avoid underflow issues in lower precision. The loss is then scaled back after the weight update.

2.5 Current Open-source Models

Multimodal Large Language Models have made significant progress in recent years, driven by advancements in transformer architectures, pre-training techniques, and multimodal integration strategies. These models are capable of processing and understanding information from multiple modalities and are increasingly being used in diverse applications such as image captioning, visual question answering, multimodal sentiment analysis, and misinformation detection. Here provides an overview of some of the most notable current MLLMs, highlighting their architectures, strengths, and areas of application. The timeline shows in Fig 3.

Vilt: The Vision-and-Language Transformer (ViLT) is a groundbreaking model designed to handle multimodal tasks that involve both visual and textual information. Introduced by researchers from Google Research in 2021, ViLT aims to bridge the gap between vision and language by directly integrating vision and language processing into a unified transformer framework. Unlike traditional multimodal models, which typically rely on separate encoders for images (e.g., CNNs) and text (e.g., transformers), ViLT uses a transformer architecture for both modalities, simplifying the model architecture and reducing computational complexity. In ViLT, images are processed by converting them into a sequence of visual tokens through a process known as patch embedding, where an image is divided into non-overlapping patches, and each patch is flattened into a vector. These patch embeddings are then passed

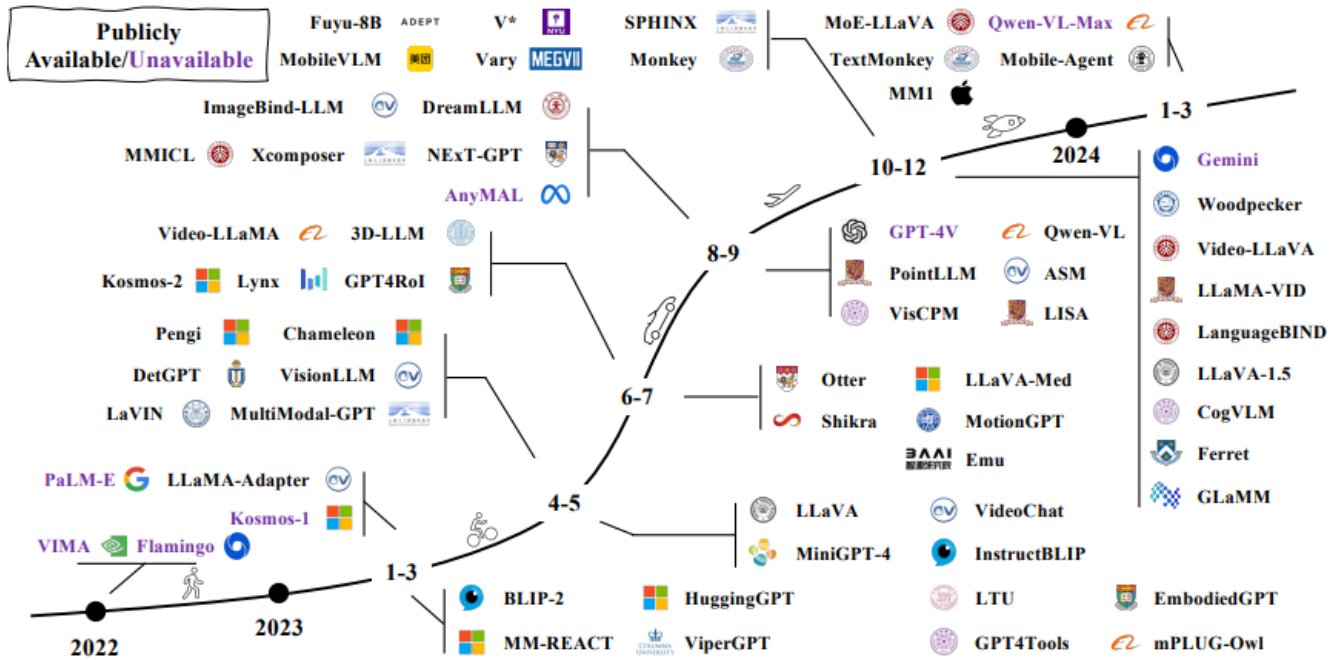


Figure 3: A timeline of representative MLLMs.(Adapted from Yin et al., 2023)

through a standard transformer model, along with text embeddings that represent the corresponding textual description or query. The transformer architecture is designed to learn joint representations of the image and text by attending to both modalities simultaneously, thus enabling cross-modal reasoning. ViLT relies on a pure transformer architecture without the need for convolutional neural networks (CNNs) for image feature extraction. This is in contrast to many previous multimodal models, such as VisualBERT or UNITER, which often use CNNs to extract image features before feeding them into the transformer model. By using a more streamlined approach, ViLT reduces the number of parameters and the computational overhead typically associated with complex image encoders, making it more efficient while still achieving high performance on a range of multimodal tasks.

in 2021 that brings together vision and language in a novel and efficient way. What sets CLIP apart from traditional models is its ability to learn from vast amounts of unlabelled data using a contrastive learning framework, allowing it to connect images and text without requiring task-specific annotations. By leveraging a combination of vision and language, CLIP achieves state-of-the-art performance on a wide range of tasks, making it a flexible tool for a variety of applications, from image classification to visual search. CLIP employs two separate encoders—one for processing images and the other for processing text. The image encoder, typically based on a convolutional neural network (CNN) or a Vision Transformer (ViT), converts images into high-dimensional vectors. The text encoder, usually based on a transformer architecture, converts textual descriptions into comparable vectors. These encoders map both modalities into a shared embedding space, where matching image-text pairs are pulled closer together, while unrelated pairs are pushed farther apart. This contrastive loss function is at the heart of CLIP's learning mechanism, enabling it to capture the underlying relationships between visual and textual representations. CLIP has great performance in zero-shot learning, which means it can tackle tasks without the need for additional training on task-specific labeled data. For example, when given a prompt like "Which image depicts a dog?", CLIP can match the correct image to the prompt based on its learned understanding of the relationship between language and vision. This zero-shot capability is enabled by CLIP's joint learning of image and text representations, making it highly adaptable to new tasks without requiring extensive retraining.

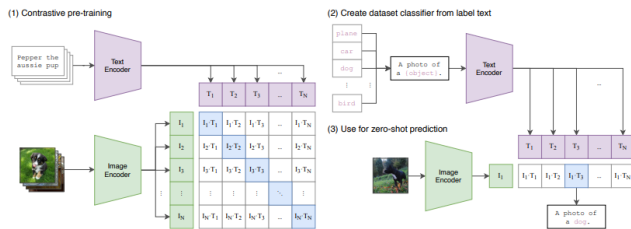


Figure 4: The Architecture of CLIP.(Adapted from Radford et al., 2021)

Clip: CLIP (Contrastive Language-Image Pre-Training) shows in Fig 4, is a pioneering multimodal model introduced by OpenAI

Qwen-VL: Qwen-VL is a part of the Qwen family of models. It specifically targets the combination of vision and textual information, enabling the model to perform tasks like image captioning, visual question answering (VQA), and cross-modal retrieval. The model is a direct extension of the powerful Qwen series, which is known for its language understanding capabilities, but with additional modifications to handle the complexities of vision-language interactions. Qwen-VL utilizes a transformer-based architecture that integrates both image and text representations into a unified model. Like other vision-and-language models, Qwen-VL processes images and text through separate encoders—one for images and one for text—but with a focus on efficiently aligning the two modalities in a shared space. The image encoder typically relies on a Vision Transformer (ViT) or similar architecture to extract visual features from the input image, while the text encoder is based on a language model (such as GPT-like models) that processes the textual input. The joint learning process enables Qwen-VL to understand and generate multimodal outputs, making it suitable for tasks that require an understanding of both visual and linguistic cues. Qwen-VL focus on fine-tuning the transformer architecture to optimize both vision and language processing. Rather than treating the two modalities separately or relying on a simple fusion of features, Qwen-VL adopts a more holistic approach to multimodal learning. The model is trained on large-scale datasets that include diverse image-text pairs, allowing it to learn robust cross-modal representations. This enables Qwen-VL to perform tasks like generating descriptive captions for images, answering questions about images, or retrieving relevant images based on a textual query.

3 Multimodal Misinformation Detection

3.1 Task introduction

Multimodal misinformation detection is a critical and rapidly evolving task in the field of artificial intelligence, aimed at identifying false or misleading content that spreads across various media types, including text, images, and videos. As misinformation becomes more sophisticated and pervasive, traditional text-based detection methods are no longer sufficient. Multimodal models, which can analyze and integrate information from different sources, offer a more robust approach to tackling this challenge. By combining textual analysis with visual content examination, these systems are better equipped to identify inconsistencies, manipulations, and contradictions within multimedia content. For instance, a misleading headline paired with an image or video that contradicts the claim can be flagged more effectively through multimodal analysis. The task requires advanced techniques in natural language processing, computer vision, and multimodal learning, allowing the model to evaluate both the textual and visual components of content simultaneously. As the spread of misinformation continues to have a significant impact on society, particularly in areas like politics, public health, and social issues, the development of accurate and efficient multimodal misinformation detection systems becomes increasingly vital to ensuring the integrity of information in the digital world.

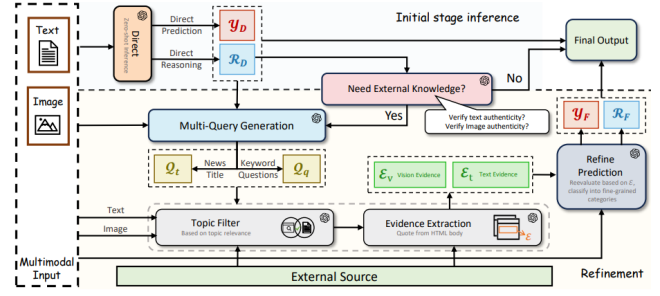


Figure 5: The pipeline of LEMMA.(Adapted from Xuan et al., 2024)

3.2 Cases studies

The detection of multimodal misinformation remains a complex challenge, necessitating the integration of diverse evidence from multiple modalities—text, image, and external knowledge—into a unified decision-making process. A variety of frameworks have been proposed to address this issue, each leveraging different aspects of multimodal reasoning and evidence retrieval to enhance the accuracy and reliability of misinformation detection.

LEMMA: Here’s a simplified English version that sounds more natural: The LEMMA framework enhances fact-checking by incorporating external knowledge into its verification process. It begins with a preliminary assessment of the image-text pair. When the initial evaluation is inconclusive, the system searches for additional context and evidence. The framework uses two approaches to gather external information: Text-based search generates a concise title from the image-text input and uses DuckDuckGo to find relevant documents; Image-based search performs a reverse image search through Google to trace the image’s origins and identify potential misuse. After collecting external documents, the system evaluates their relevance and selects the most pertinent ones for analysis. Key information, including main content, titles, and publication dates, is extracted and used as supporting evidence. The final verdict combines both the initial assessment and this external knowledge. The framework is tested on Twitter and Fakeddit datasets has shown that this approach significantly improves detection accuracy by supplementing the model’s reasoning with real-world contextual information.

FKA-Owl: The FKA-Owl framework improves image authenticity detection through an innovative knowledge fusion approach. It utilizes the ImageBind-Huge model for processing visual features while employing a dedicated text encoder for semantic information extraction. Through cross-attention mechanisms, the framework effectively links image and text information. For detecting image manipulation traces, FKA-Owl uses deconvolution layer techniques to convert low-resolution features into high-resolution ones, better preserving image details. The framework employs dual-branch feature extraction to integrate text and visual features into a unified space for analysis. Finally, the framework combines multiple dimensions of information - including image semantics, textual content, and manipulation traces - to form a comprehensive judgment. Testing on the DGM4 and NewsCLIPPings datasets demonstrates

the framework’s excellent performance across various types of information detection.

SNIFFER: The SNIFFER framework addresses out-of-context (OOC) misinformation detection, where text and images are mismatched. It improves LVLM performance by adding external knowledge and instruction-based fine-tuning. SNIFFER uses ViT-G/14 to extract visual features and Vicuna-13B for text processing. The Q-Former module converts image features into soft prompts, guiding the LVLM with textual instructions for reasoning. The framework has two fine-tuning phases. First, the model is fine-tuned on NewsCLippings data to align image descriptions with detailed entities. In the second phase, OOC instructions teach the model to spot and explain discrepancies between text and images. This improves interpretability and transparency. The model’s reasoning is further enhanced by using the Google Entity Detection API to extract entities from images and supplementing fine-tuning with this knowledge. Reverse image search is also used to find relevant web pages, checking the relationship between the image and text. The final prediction is made after consistency checks. Evaluations on the NewsCLippings dataset show improved detection and explainability of OOC misinformation.

3.3 Strengths and Limitations of MLLMs

The use of Multimodal Large Language Models (MLLMs) in detecting multimodal misinformation has demonstrated substantial advancements in the accuracy and robustness of fake news detection. These models integrate various modalities, and external knowledge, to leverage the complementary strengths of each modality.

MLLMs excel at combining the reasoning capabilities of large language models (LLMs) with the rich information encoded in images and external knowledge sources. For instance, frameworks like LEMMA and FKA-Owl utilize external knowledge augmentation, enabling models to perform more complex reasoning by consulting relevant databases or using external queries. This ability enhances the detection of nuanced misinformation, especially when direct evidence from the input data (text or image) is insufficient. Such models can determine when additional context is needed, offering a more informed judgment than traditional models.

MLLMs, designed for cross-domain misinformation detection, show a remarkable ability to generalize across different types of content. This is important given that fake news often spans a variety of topics and domains. By using cross-modal attention and external knowledge, such models can transfer knowledge from one domain to another, improving performance on diverse datasets such as DGM4, NewsCLippings, and Twitter.

MLLMs have ability to process and integrate different modalities of data. For example, in approaches like FKA-Owl, image features and textual features are aligned using advanced techniques like cross-modal attention. This multi-layered integration allows the models to effectively capture the semantic relationships between text and images, which is crucial for understanding multimodal misinformation that often relies on mismatches between textual and visual content.

The models discussed above often operate in environments where input data can be noisy or adversarially manipulated. For instance, images may be subtly altered to deceive the model, and

the text may include sarcastic, ironic, or ambiguous statements that are difficult for models to interpret. While some frameworks, like SNIFFER, attempt to address this with techniques such as internal consistency checks and external verification, noisy data and adversarial inputs still pose a significant challenge for MLLMs. This is particularly true when the misinformation is not explicit but instead relies on more nuanced or indirect cues, which are harder for models to detect.

The integration of multiple modalities and external knowledge retrieval significantly increases the complexity of the models. These systems often require powerful computational resources, both for training and inference, which may limit their accessibility and scalability. For example, models like FKA-Owl that involve multi-level visual features and advanced reasoning layers demand substantial GPU capabilities and large-scale datasets, making them expensive to deploy and maintain, especially for smaller organizations or research groups.

4 Conclusion

This paper has provided a comprehensive examination of Multimodal Large Language Models (MLLMs) with a particular focus on their application in misinformation detection. Through our analysis, we have explored the fundamental architecture of MLLMs, including transformer components, pre-training strategies, and efficiency optimizations that make these complex models more practical for real-world applications. Our review of current open-source models such as Vilt, CLIP, and Qwen-VL demonstrates the rapid advancement and diversification of approaches in multimodal processing.

In examining the specific application of MLLMs to misinformation detection, we have highlighted both the potential and challenges of these systems. The emergence of specialized datasets like MMFakeBench, VERITE, and M3A has provided crucial resources for training and evaluating multimodal misinformation detection systems. Our case studies of frameworks such as LEMMA, FKA-Owl, and SNIFFER illustrate the various approaches to integrating multiple modalities and external knowledge for more accurate detection of false information.

While MLLMs show promising results in combining text, image, and external knowledge for sophisticated reasoning tasks, several challenges remain. These include the dependency on high-quality external data sources, vulnerability to adversarial attacks, and the substantial computational resources required for deployment. Future research should focus on addressing these limitations while maintaining the models’ strong performance in cross-modal understanding and reasoning.

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